

Homework Exercise 3

Angelica Ortiz 11191225, Daniel Nägele 11180610

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1 Exercise 3 – RFID

The RFID system is based on Dynamic Frame ALOHA and contains $N = 4$ tags.

The collision resolution efficiency is defined as:

$$\eta = \frac{N}{E[T]} = \frac{4}{E[T]}$$

The expected resolution time for each initial frame size r_1 is:

$$E[T] = r_1 + \sum_{i=0}^3 P(S = i) L_{4-i}$$

where S is the number of successfully transmitted tags in the first frame. Using the given values:

$$L_2 = 4, \quad L_3 = \frac{51}{8} = 6.375$$

and computing L_4 for $r = N = 4$:

$$L_4 = \frac{4 + 0.1875L_3 + 0.5625L_2}{1 - 0.15625}$$

$$L_4 = \frac{4 + 0.1875 \cdot 6.375 + 0.5625 \cdot 4}{0.84375} = 8.8241$$

The resulting efficiencies are:

r_1	$E[T]$	η
1	9.8241	0.4072
2	9.5995	0.4167
3	8.9544	0.4467
4	8.8241	0.4533
5	9.0377	0.4426
6	9.4661	0.4226

The maximum efficiency is obtained for:

$$r_1 = 4$$

with:

$$\eta_{\max} \approx 0.4533$$

This result is expected because the optimal initial frame size is equal to the number of tags, $r_1 = N = 4$. In this case, the probability of having one tag per slot is maximized, reducing both collisions and empty slots.

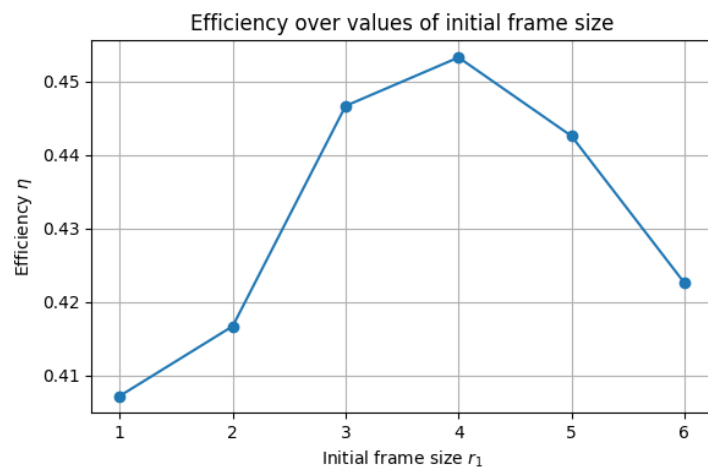


Figure 1: Efficiency over values of initial frame size